HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
School of Information and Communications Technology

Paris 2024 Olympic Data Story

A Streamlit dashboard and its visualization techniques

IT5425 – Data Management & Visualization · Capstone demo

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Dataset: Paris 2024 public CSVs (GitHub paris2024-data + data.gouv.fr, ODbL) · Tool: Streamlit + Plotly

The dataset

Eight public Paris 2024 CSVs, joined into analysis-ready tables and findable via Google Dataset Search:

- 11,183 athletes across 206 NOCs
- 45 disciplines, 2,271 athlete–medal records
- Enriched athletes: age, venue coordinates, dates, record flag
- 41 venues with valid coordinates

Quirks handled in code: semicolon + French + BOM enriched file, an unnamed index column, NOC→ISO-3 mapping, French discipline/gender translation.

Two distinct medal concepts

Country totals count official medals by NOC.

Athlete–medal records count each athlete on a medal, so team sports multiply.

The dashboard keeps them on separate pages and labels the unit.

Objective and story arc

Problem. A flat medal table hides geography, specialization, demographics, and the spatial footprint of the Games.

Goal. A guided, interactive story that pairs explanatory narrative with exploratory filtering — four linked sections, broad to narrow:

1	2	3	4
Opening Lens	Medal Race	Specialization	Athlete Lens
global scale, geography & venues	led & moved vs Tokyo	NOC × discipline	age, gender, load, records

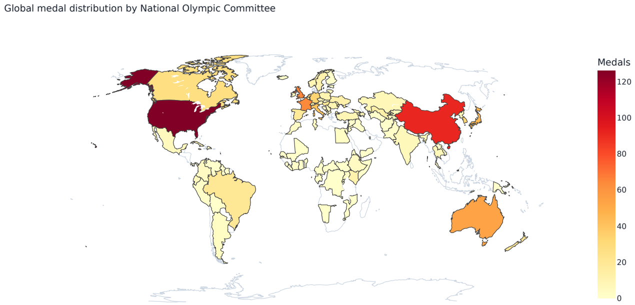
Headline

USA and China tied at 40 golds (USA led totals 126–91); host France +31 vs Tokyo, Japan −13; 49.1% female; median age 26; 64 records.

Which technique each chart applies

Chart	Technique demonstrated
World choropleth	Map intensity; nominal→region, quantitative→colour
Stacked / diverging bars	Length for amounts; diverging bar for change
Bubble scatter	Relationship; position + area + hue; overlap control
Country×discipline heatmap	Pre-attentive detection + exact cell values
Medal-flow Sankey	Bipartite graph (NOC→discipline); ribbon width = flow
Treemap	Hierarchy with proportional ink
Histogram / box plots	Distributions and spread
Venue symbol map	Point map; size + colour on coordinates
Daily medal timeline	Trend over time; animated cumulative race
Inline controls	Filtering, view transform, animation, details-on-demand

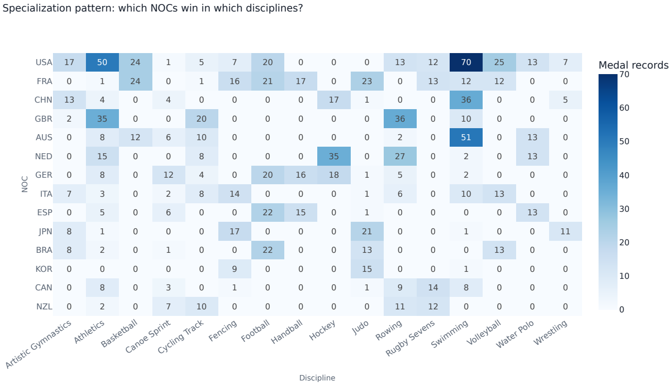
Opening Lens — global medal geography



Colour intensity makes the USA–China lead pop pre-attentively before any reading. Olympic NOC codes are mapped to ISO-3 so geometry resolves.

Technique: Choropleth; colour-intensity encoding; pre-attentive processing.

Sport Specialization — who wins what

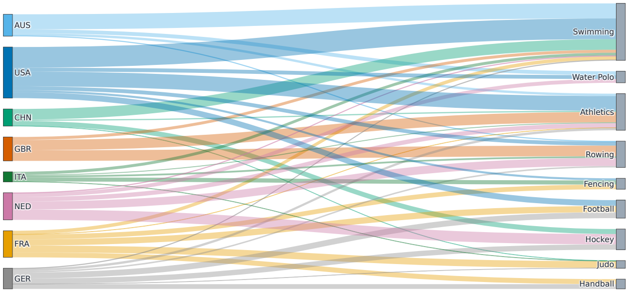


Dark cells (USA swimming =70, NED hockey, GBR rowing) are spotted instantly; printed values give exact magnitudes where colour is imprecise.

Technique: Pre-attentive detection + multiple encodings.

A derived flow graph

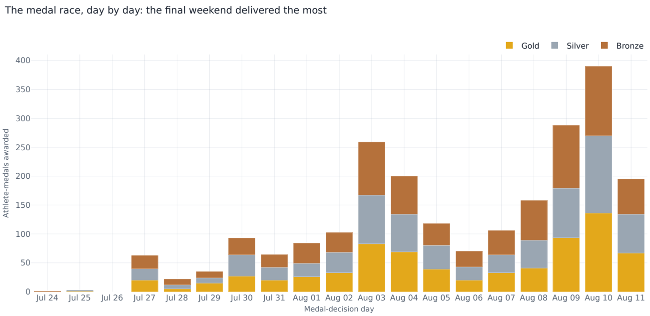
NOC to discipline: where each nation's medals come from



A bipartite flow, NOC → discipline, with ribbon width = record volume. The widest ribbons mark the strongest nation–sport pairings.

Technique: Graph visualization (Sankey mechanism).

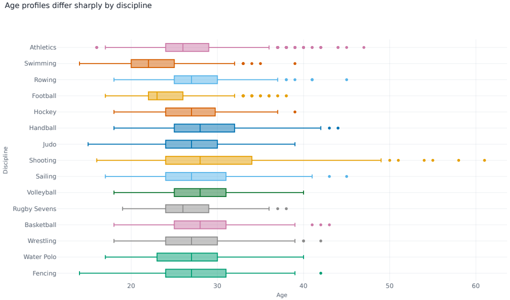
The medal race over time — and the one animation

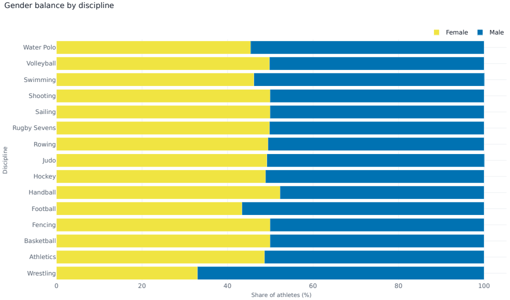


Athlete-medals by Games day: the tally surges on the closing weekend. A Daily / Cumulative toggle and an animated bar-chart race (press play) add the time dimension.

Technique: Trend over time; animation (used only where the variable is time-ordered).

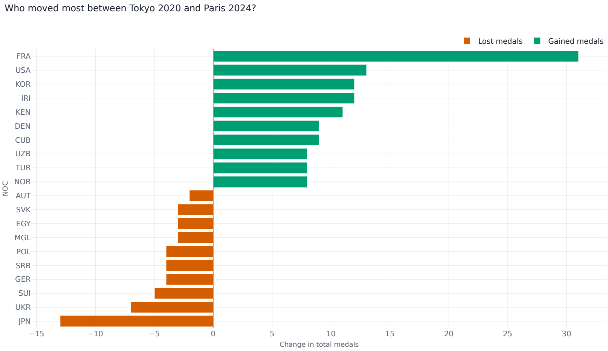
Athlete Lens — distributions and proportions





Age medians span 20 (skateboarding) to 39 (equestrian); near gender parity at 49.1% female. Technique: Distributions (box) & proportions (100% stacked).

Medal Race — who moved most since Tokyo

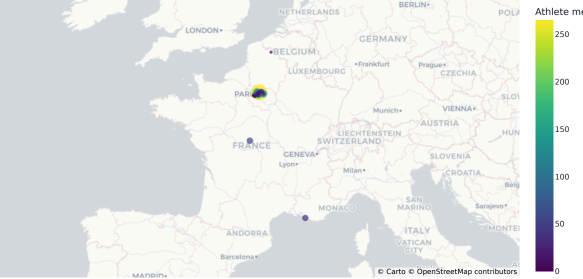


A diverging bar around zero: host France +31, while Japan −13. Direction is labelled so colour is never the only cue.

Technique: Visualizing change; accessible colour.

Opening Lens — the spatial footprint

Venue geography: where athletes competed across France



A symbol map closing the Opening Lens: bubble size = athlete volume, colour = medal activity. The dense Paris cluster contrasts with sailing/surf venues far to the south.

Technique: Point map on lat/long coordinates.

Interaction and storytelling

Interaction

- No sidebar — tabs switch sections; controls sit beside each chart
- Inline segmented pills + multiselects + sliders
- Details-on-demand: hover tooltips; zoom/pan on every chart
- Empty filters fall back to styled placeholders, never crash

Storytelling

- Author-driven arc, reader-driven on each page
- Each section leads with headline KPI cards, then answers visually
- No raw data dumps — show the data, detail on hover
- Colourblind-safe Okabe–Ito palette; host-nation spotlight

Where we drew the line

Animation is used — but only on the time-ordered medal race, where motion carries information. Uncertainty bands and trend smoothing are left out: a completed-event snapshot has no sampling error to show.

Key insights

1. Gold was a dead heat, totals were not. USA & China each 40 gold; USA led totals 126–91 through breadth.
2. Hosting moved the table. France +31 vs Tokyo (largest swing); Japan −13.
3. Specialization defines mid-tier powers. NED hockey, GBR rowing, AUS swimming — concentrated, not broad.
4. Near gender parity, sport-specific ages. 49.1% female; medians from 20 to 39 by discipline.
5. A dispersed Games. Venues reach from Paris to Marseille sailing and overseas surfing.

Reproducible by design

Figures match the dashboard

- Every figure is generated from the same code that drives the live dashboard
- No screenshots and no hand-drawn charts — the report cannot drift from the app
- A single pipeline regenerates all figures from the source data

Checked

- All four sections and every control exercised, including empty and narrow selections
- Empty selections degrade gracefully to a styled “no data” state
- Row counts, data types, and missing-data rates reconcile with the source

Thank you Questions welcome.